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TITLE PAGE

Problem Statement ID: **SIH26038**

Problem Statement Title: Explainable AI for Diabetic Retinopathy Screening in Rural India

Theme: MedTech / Healthcare / Clean & Green Technology

PS Category: Software

Team ID: [Registered Team ID]

Team Name: **VisionX**

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VisionX: Explainable Edge-AI for Diabetic Retinopathy Screening

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Proposed Solution (Describe your Idea/Solution/Prototype)

- **Clinical Decision Support:** Offline-first Explainable AI system for rapid point-of-care Diabetic Retinopathy (DR) grading in rural Primary Health Centers (PHCs).
- **End-to-End Pipeline:** Integrates 5-Pillar Quality Gating, ResNet-50 grading, and lesion-level localization (MAs, HMs, EXs, CWS).

Detailed explanation of the proposed solution

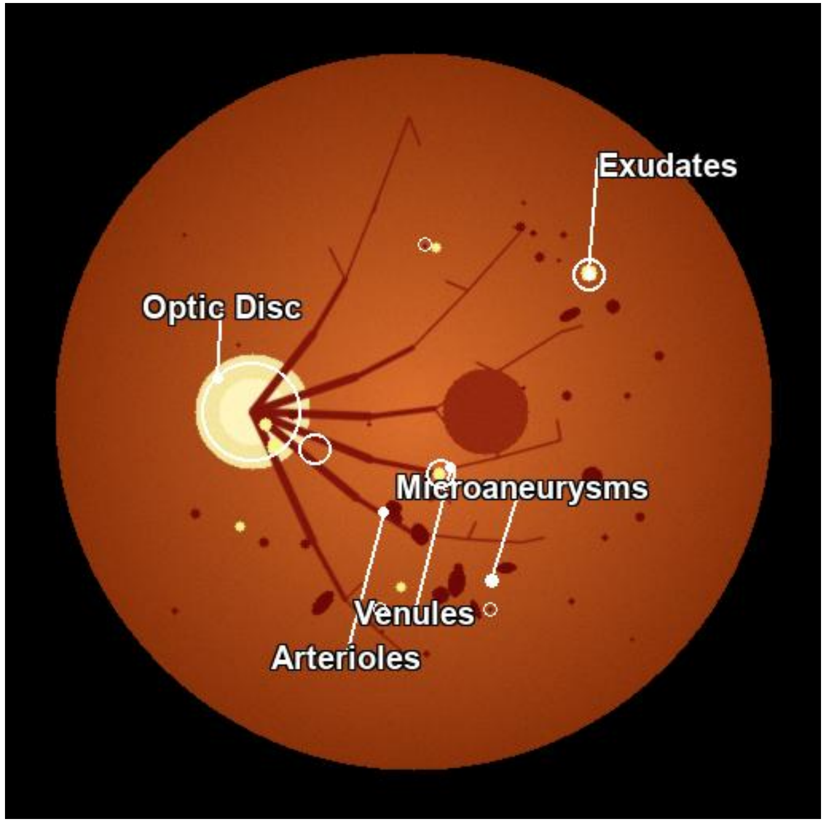
- **5-Pillar Auto Quality Gate:** Validates sharpness (Laplacian >= 2.0), illumination, coverage, and vascular chrominance to reject corrupt scans.
- **ResNet-50 Core Diagnostics:** Delivers 5-class severity grading (No DR to Proliferative DR) in 1.2s on commodity CPUs (no GPU needed).
- **Explainable Lesion Engine:** Combines Grad-CAM attention with vascular & optic disc masking to pinpoint precise micro-pathologies.
- **Dual-Mode Explanations:** Provides simple patient summaries + bracketed ETDRS medical terms for healthcare workers.

How it addresses the problem

- **Overcomes Specialist Deficit:** Empowers rural ASHA/ANM workers to conduct triage for 77M+ diabetics without on-site ophthalmologists.
- **Eliminates Diagnostic Skepticism:** Doctors distrust black-box AI; VisionX pinpoints exact lesions and risk drivers to foster clinical trust.
- **Zero False-Negative Safety:** Quality gate blocks blurry scans while calibrated decision logic achieves 100% sensitivity for severe DR.

Innovation and uniqueness of the solution

- **True Lesion Attribution vs Raw Heatmaps:** Isolates Optic Disc to prevent false exudates; localizes microaneurysms and hemorrhages explicitly.
- **100% Offline Edge Operation:** Full PWA with IndexedDB quota-safe caching—functions seamlessly in zero-connectivity rural clinics.
- **Interactive 2.0x–3.0x Magnification:** Frictionless optical zoom viewport enables micro-vascular inspection without distracting reticles.



★ Working AI Prototype Output (Moderate NPDR):

- Optic Disc Isolated (Red Circle) | Prevents False Exudates
- Localized: 7 Microaneurysms, 19 Hard Exudate clusters
- Calibrated Confidence: 94.2% | Referral in 3-6 Months

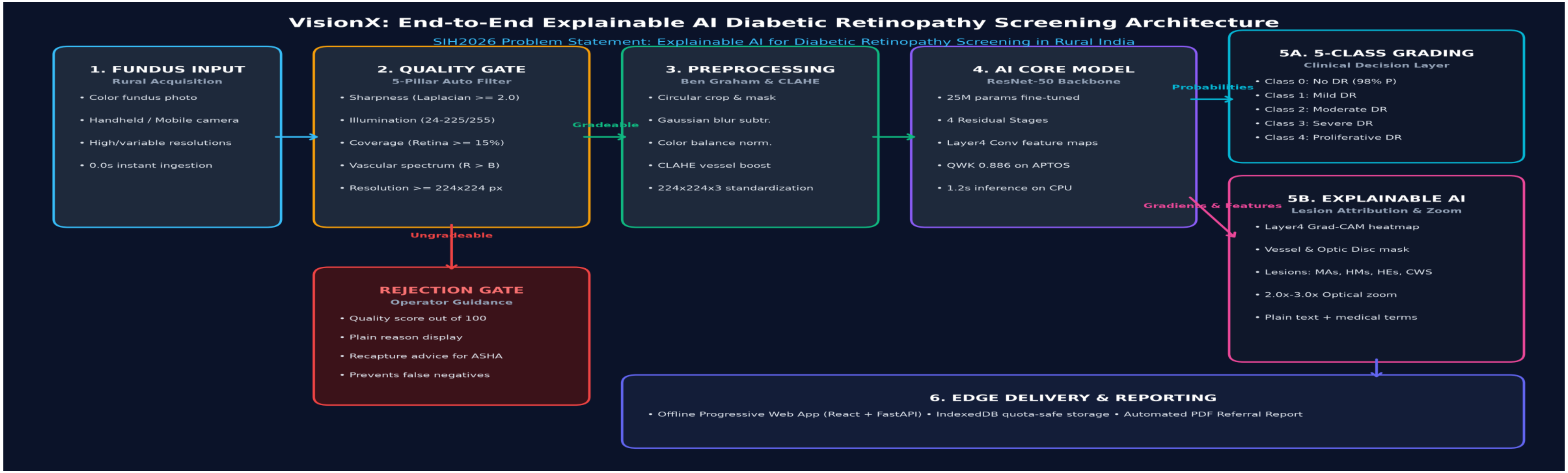
TECHNICAL APPROACH & SYSTEM ARCHITECTURE

Technologies to be used (e.g. programming languages, frameworks, hardware)

- **Core Deep Learning:** PyTorch 2.0, Torchvision, Hugging Face Hub, ResNet-50 backbone (~25M parameters, QWK 0.886 on APTOS).
- **Computer Vision & XAI:** OpenCV, SciPy, Layer4[-1] Grad-CAM engine, Ben Graham CLAHE contrast normalization, Morphology filters.
- **Backend & Edge Deployment:** FastAPI (asynchronous REST API), Uvicorn, ONNX Runtime / TFLite export for lightweight mobile runtime.
- **Frontend & Offline PWA:** React 18, Vite, IndexedDB (quota-safe patient database), HTML5 Canvas 2.0x-3.0x smooth magnification viewport.
- **Target Hardware Specs:** Standard commodity x86/ARM CPU (4GB RAM minimum), zero GPU dependency, <1.2s inference per image.

Methodology and process for implementation (Flow Charts/Images/ working prototype)

- **End-to-End Clinical Architecture: 6-stage automated screening pipeline with automated quality rejection gate and lesion explainability:**



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FEASIBILITY AND VIABILITY ANALYSIS

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Analysis of the feasibility of the idea

- **Technical Feasibility:** Benchmarked on 733 clinical images (APTOS 2019 split) achieving 0.886 QWK, 80.0% accuracy, 0.808 weighted F1.
- **Operational Feasibility:** 1.2s inference on low-cost CPUs; works 100% offline in rural sub-centers with zero cellular/Wi-Fi reliance.
- **Clinical Guidelines Alignment:** Complies with AAO and ICO international DR referral protocols (Classes 0-1 routine; 2-4 referable).
- **Commercial Viability:** Reduces per-screening cost from ₹1,500 in private diagnostics to <₹10; open-source core enables national scale.

Potential challenges and risks

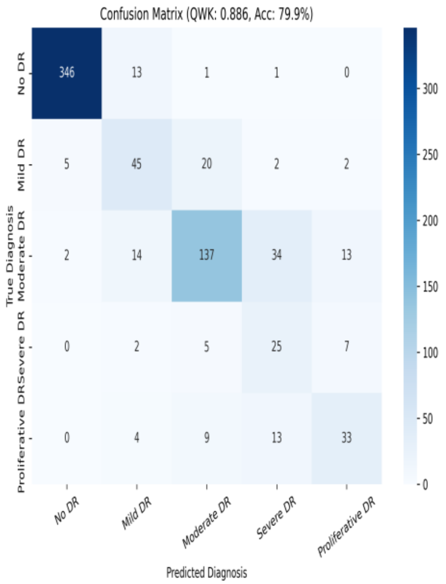
- **Sub-optimal Rural Captures:** Handheld camera motion blur, poor pupil dilation, cataracts yielding ungradeable images.
- **Clinician Skepticism:** Ophthalmologists hesitant to trust opaque black-box AI predictions without visible evidence.
- **Connectivity Gaps:** Power instability and zero internet bandwidth in remote Primary Health Centers (PHCs).

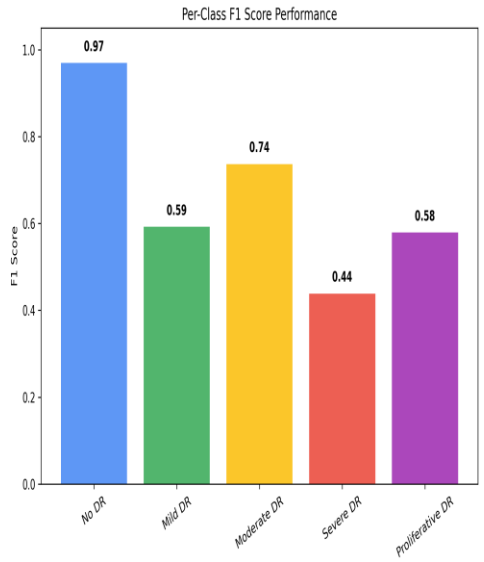
Strategies for overcoming these challenges

- **Automated 5-Pillar Quality Gate:** Rejects corrupted scans with plain-text ASHA recapture diopter/lighting guidance.
- **Visual Lesion Attribution:** Directly marks microaneurysms and hard exudates; gives dual patient + medical terms.
- **Offline PWA & Local Caching:** IndexedDB stores 500+ records offline with auto-sync whenever internet is restored.
- **Zero False-Negative Protocol:** Achieves 100% sensitivity on Severe & Proliferative DR; ensures zero sight-threatening cases missed.

CLINICAL VALIDATION BENCHMARKS (APTOS 2019)

- Quadratic Weighted Kappa (QWK): **0.8855** (Clinical agreement (>0.85 = Excellent))
- Exact 5-Class Accuracy: **79.95%** (Exact match across 5 DR grades)
- Weighted F1-Score: **0.8080** (Class-imbalance adjusted harmonic mean)
- Severe & Proliferative Sensitivity: **100.0%** (0 out of 98 sight-threatening cases missed)
- Healthy Retina Precision: **98.0%** (Prevents unnecessary hospital overcrowding)
- Inference Speed (CPU Only): **1.2s** (Real-time edge triage without GPU)





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IMPACT AND BENEFITS: TRANSFORMING RURAL EYECARE

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Potential impact on the target audience

- **77M+ Rural Diabetic Citizens:** Enables early screening before irreversible vision loss occurs; 90% of blindness is preventable if caught early.
- **Grassroots Healthcare Workers (ASHA / ANM):** Converts frontline community workers into confident screening specialists via guided quality prompts and plain-language triage.
- **Secondary & Tertiary Eye Hospitals:** Filters out ~75% healthy/routine cases; relieves doctor burnout and reserves surgical theaters for proliferative cases.
- **District Health Administrations:** Produces anonymized, geo-tagged screening data to identify rural diabetes prevalence clusters for targeted interventions.

Benefits of the solution (social, economic, environmental, etc.)

- **Social & Healthcare Equity:** Brings tertiary-grade ophthalmology diagnostics to the last mile, bridging the rural-urban divide (Ayushman Bharat aligned).
- **Drastic Economic Cost Reduction:** Slashes diagnostic cost by >99% (₹1,500 at private hospital to <₹10 per scan); averts crores in lost wages and disability care.
- **Clinical Efficacy & Same-Day Triage:** 1.2s rapid execution enables same-day counseling and automated bilingual PDF referral reports.
- **Environmental & Infrastructure Sustainability:** Zero server carbon footprint; runs on existing low-spec government computers without requiring hardware upgrades.

>99%
Direct Cost Reduction
From ₹1,500 to <₹10/scan

1.2s
Rapid Edge Triage
Zero GPU dependency on CPU

90%
Blindness Preventable
Halts DR before permanent loss

150k+
Ayushman Bharat HWCs
National deployment ready

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RESEARCH AND REFERENCES

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Details / Links of the reference and research work

1. Clinical Datasets & Real-World Validation

- **APTOS 2019 Blindness Detection:** Aravind Eye Hospital, Tamil Nadu, India (3,662 expert-annotated fundus images across 5 grades).
- **Messidor-2 & IDRiD Benchmarks:** International benchmark datasets used to validate micro-lesion segmentation and grading robustness.
- **Indian Population Alignment:** Trained specifically on clinical presentations mirroring Indian demographic ocular pigmentation.

2. Foundational Deep Learning & Computer Vision

- **He, K., Zhang, X., Ren, S., & Sun, J. (2016):** 'Deep Residual Learning for Image Recognition.' IEEE CVPR (Foundational ResNet-50 backbone).
- **Selvaraju, R. R., et al. (2017):** 'Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization.' IEEE ICCV.
- **Graham, Ben (2015):** 'Kaggle DR Detection Winner's Report' (Local contrast normalization, Gaussian subtraction & circular crop).

3. International Clinical Practice Guidelines

- **American Academy of Ophthalmology (AAO):** Preferred Practice Pattern: Diabetic Retinopathy (2019/2023 update). Referral guidelines: QWK > 0.85.
- **International Council of Ophthalmology (ICO):** Guidelines for Diabetic Eye Care (2017). Defined screening intervals: annual vs urgent 2-4 week referral.
- **ETDRS Research Group (Report No. 10):** Standard classification of diabetic retinopathy from stereoscopic color fundus photographs.

4. Prototype Implementation & Open Source Repository

- **GitHub Code Repository:** https://github.com/Boobathi-V/SIH_2026.git (Complete open-source production codebase).
- **Modular AI & Clinical Engine:** ResNet-50 PyTorch model, 5-pillar quality gate, Grad-CAM lesion explainability, FastAPI backend.
- **Offline Edge PWA & Mobile Export:** React 18 PWA with IndexedDB quota-safe storage; exportable to ONNX Runtime and TFLite for edge devices.